

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: ITA 06

COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO3: Bayes Theorem – Concept Learning – Maximum Likelihood – Minimum Description Length Principle – Bayes Optimal Classifier – Gibbs Algorithm – Naïve Bayes Classifier – Bayesian Belief Network – EM Algorithm – Probability Learning – Sample Complexity – Finite and Infinite Hypothesis Spaces – Mistake Bound Model, (BL 6)

Assessment Tool Weightage: 20%

QUESTIONS: 15

Total Marks:25

Annexure A

Case Study

Code of Conduct:

I Pravin R / 192421292 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Annexure A

Short Answer Test

1, A hospital aims to develop an intelligent system to diagnose diseases based on patient symptoms and medical history. Using concepts such as Bayes' Theorem, Naïve Bayes classifiers, and Bayesian Belief Networks, design a probabilistic model for diagnosis. Analyze how uncertainty is handled and how conditional dependencies between symptoms influence predictions. Discuss the role of probability learning and evaluate the effectiveness of your approach with respect to real-world constraints.

2. An organization wants to build a spam detection system for emails using machine learning techniques. Using Naïve Bayes and Bayes Optimal Classifier concepts, design a classification system. Compare Maximum Likelihood estimation with Minimum Description Length (MDL) principle in feature selection. Analyze system performance, discuss trade-offs in model complexity, and evaluate how sample size and hypothesis space affect classification accuracy.

3. A telecommunications company aims to predict customer churn using historical usage data and service feedback. Using probabilistic models such as **Naïve Bayes**, **Bayesian inference**, and hypothesis space search, design a predictive system. Analyze how prior probabilities and feature dependencies influence churn prediction. Evaluate how uncertainty in customer behavior is handled and discuss the effectiveness of the model in improving customer retention strategies.

4. A smart healthcare system is designed to monitor patients in real-time using wearable sensors. Using **Bayesian Belief Networks** and probability learning, develop a model to detect anomalies in vital signs. Analyze how conditional dependencies between physiological parameters are modeled and how uncertainty due to sensor noise is handled. Evaluate system performance and discuss challenges in real-time deployment.

5. A logistics company wants to optimize delivery routes under uncertain traffic and weather conditions. Using **Bayesian inference** and probabilistic reasoning, design a system that predicts optimal routes dynamically. Analyze how prior knowledge and real-time data updates influence decision-making. Evaluate computational complexity and discuss how uncertainty impacts route optimization efficiency.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	20	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	15	Clear, concise, and well-organized answer	Understandable with minor issues	Somewhat unclear or poorly structured	Difficult to understand

1. Intelligent Disease Diagnosis Using Bayesian Models

Proposed System

A hospital can build a probabilistic diagnosis system where diseases are treated as hypotheses and patient symptoms/medical history are evidence.

Using Bayes' Theorem:

$$P(D|S) = \frac{P(S|D)P(D)}{P(S)}$$

Where:

D = disease

S = observed symptoms

P(D) = prior probability of disease

P(S|D) = probability of symptoms given the disease

P(D|S) = probability of disease given symptoms

Naïve Bayes Model

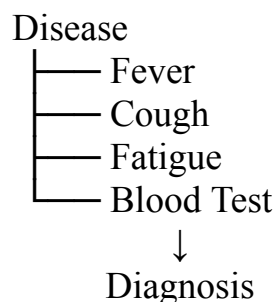
For multiple symptoms:

$$P(D|S_1, S_2, \dots, S_n) \propto P(D) \prod_{i=1}^n P(S_i|D)$$

For example, if a patient has fever, cough, and fatigue, the system calculates the probability of diseases such as influenza, pneumonia, or COVID-19 and selects the disease with the highest posterior probability.

Bayesian Belief Network

A Bayesian Belief Network (BBN) can represent dependencies between variables:



Unlike Naïve Bayes, a BBN can model relationships between symptoms. For example, fever may influence heart rate, while both can be influenced by the underlying disease.

Handling Uncertainty

Medical diagnosis is uncertain because:

1. Symptoms can occur in multiple diseases.
2. Patients may have incomplete medical histories.
3. Sensor/test results may contain errors.
4. Different patients respond differently to the same disease.

Probabilities allow the system to express uncertainty instead of producing only a yes/no answer.

Probability Learning

The probabilities can be learned from historical patient records using Maximum Likelihood Estimation (MLE) or Bayesian parameter estimation.

Evaluation

Advantages:

1. Handles incomplete information.
2. Produces probability-based predictions.
3. Can incorporate medical expert knowledge.
4. BBNs represent dependencies between medical variables.

Limitations:

1. Requires reliable medical datasets.
2. Incorrect probabilities can produce incorrect diagnoses.
3. Patient privacy is a major concern.
4. Should assist doctors rather than replace clinical judgment.
5. Conclusion: A combination of Bayesian inference and BBNs can provide an effective decision-support system, especially when uncertainty and incomplete information are common.

2. Email Spam Detection Using Naïve Bayes and Bayes Optimal Classifier

Proposed System

The system classifies an email as:

1. Spam
2. Not Spam

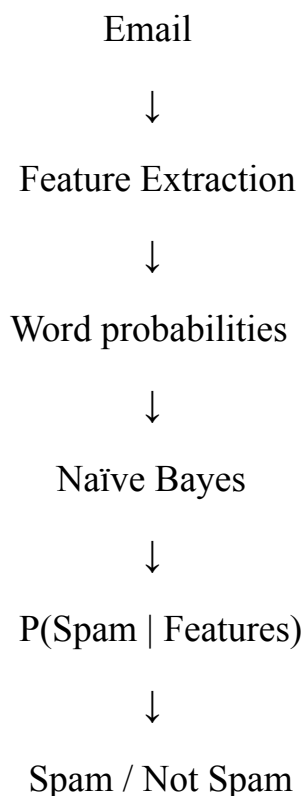
Important features include:

1. Words such as "free", "offer", "winner"
2. Number of links
3. Number of attachments
4. Sender information
5. Email length
6. Presence of suspicious phrases

Naïve Bayes

$$P(\text{Class} | \text{Features}) \propto P(\text{Class}) \prod_i P(\text{Feature}_i | \text{Class})$$

For example:



If:

$$P(\text{Spam} | \text{Email}) > P(\text{NotSpam} | \text{Email})$$

then the email is classified as spam.

Bayes Optimal Classifier

Instead of selecting only one hypothesis, the Bayes Optimal Classifier considers all possible hypotheses:

$$P(y | x) = \sum_h P(y | x, h) P(h | x)$$

It can theoretically provide better predictions because uncertainty among different hypotheses is considered.

MLE vs MDL

MLE	MDL
Maximum Likelihood Estimation	Minimum Description Length
Selects parameters that maximize likelihood	Selects a model that gives the shortest total description
Focuses on fitting training data	Balances model complexity and data fit
Can overfit with many features	Helps reduce overfitting
Usually simpler	More computationally demanding

Sample Size and Hypothesis Space

With a small dataset, the probability estimates may be unreliable.

With a large dataset, the system gets better estimates of:

$$P(\text{word} | \text{spam})$$

A very large hypothesis space can increase the possibility of overfitting.

Performance Evaluation

Use:

1. Accuracy
2. Precision
3. Recall
4. F1-score
5. Confusion matrix

For spam detection, precision and recall are especially important because incorrectly classifying a legitimate email as spam can be costly.

Conclusion

Naïve Bayes is fast and works well for text classification. Bayes Optimal classification can theoretically provide more robust predictions but may require more computation. MDL helps control model complexity, while sufficient training data improves probability estimation.

3. Customer Churn Prediction Using Probabilistic Models

Proposed System

The telecommunications company can predict whether a customer will:

1. Churn
2. Stay

Features can include:

1. Monthly bill
2. Call duration
3. Internet usage
4. Number of complaints
5. Contract type
6. Customer age
7. Service quality
8. Customer feedback

Naïve Bayes

The system calculates:

$$P(\text{Churn} \mid \text{Features}) \propto P(\text{Churn}) \prod_i P(\text{Feature}_i \mid \text{Churn})$$

For example:

Usage Data + Feedback



Feature Extraction



Naïve Bayes



$P(\text{Churn} \mid \text{Customer Data})$



High / Medium / Low Risk

Prior Probability

Suppose historically:

$P(\text{Churn})=0.20$

and:

$P(\text{Stay})=0.80$

A new customer's features update these probabilities through Bayesian inference.

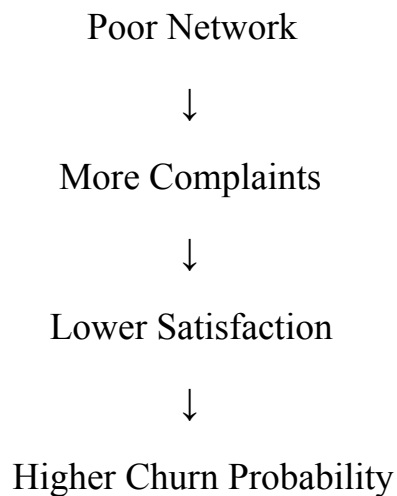
If a customer has many complaints and declining usage, the posterior churn probability may increase significantly.

Feature Dependencies

Naïve Bayes assumes features are conditionally independent:

$P(X_1, X_2 \mid C) = P(X_1 \mid C)P(X_2 \mid C)$

However, real-world variables may be related. For example:



A Bayesian network can represent these dependencies more realistically.

Hypothesis Space Search

The system can consider different possible models/hypotheses and select one that best explains historical customer behavior.

Handling Uncertainty

Customer behavior is unpredictable. Bayesian inference represents this uncertainty using probabilities rather than fixed predictions.

For example:

$$P(\text{Churn})=0.82$$

means the customer has a high estimated churn risk, not that churn is guaranteed.

Retention Strategy

The company can prioritize high-risk customers and provide:

1. Special offers
2. Better service plans
3. Technical support
4. Personalized discounts

Conclusion

A Bayesian churn prediction system can help organizations identify high-risk customers early. Naïve Bayes provides a simple and fast solution, while Bayesian networks can better represent dependencies between customer behaviors.

4. Real-Time Healthcare Monitoring Using Bayesian Belief Networks

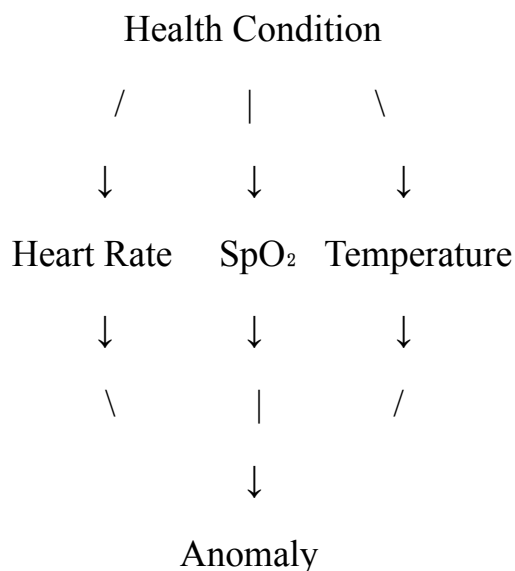
Proposed System

Wearable sensors can continuously collect:

1. Heart rate
2. Blood pressure
3. SpO₂
4. Body temperature
5. Respiratory rate
6. ECG data

A Bayesian Belief Network can model relationships between these parameters.

Example BBN



For example, a health condition may influence heart rate and oxygen saturation simultaneously.

Probability Learning

Historical patient data can be used to learn:

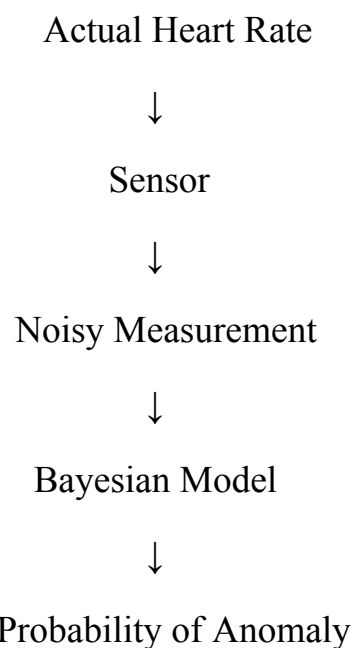
$P(\text{HeartRate} \mid \text{HealthCondition})$ $P(\text{SpO2} \mid \text{HealthCondition})$

and other conditional probabilities.

Handling Sensor Noise

Wearable sensors are not perfectly accurate.

For example:



Instead of assuming every abnormal measurement is a real medical problem, the Bayesian system considers the probability that the measurement is caused by noise.

Example

If a sensor suddenly reports:

HeartRate=150

the system should not immediately declare an emergency.

It can combine:

1. Previous heart rate
2. Current heart rate

3. SpO₂
4. Blood pressure
5. Patient condition

to calculate:

$P(\text{Anomaly} \mid \text{SensorData})$

Performance Evaluation

Metrics include:

1. Detection accuracy
2. False-positive rate
3. False-negative rate
4. Detection latency
5. Sensitivity
6. Specificity

Real-Time Challenges

1. Sensor noise
2. Missing data
3. Battery limitations
4. Network latency
5. Computational resources
6. Patient privacy
7. False alarms

Conclusion

Bayesian Belief Networks are suitable because they can model relationships between physiological parameters and handle uncertain sensor measurements. Real-time deployment requires efficient computation and reliable sensor data.

5. Dynamic Route Optimization Under Traffic and Weather Uncertainty

Proposed System

A logistics company can combine historical information with real-time data to dynamically select delivery routes.

Input data:

1. Historical traffic
2. Current traffic
3. Weather
4. Road conditions
5. Distance
6. Delivery deadlines
7. Accident reports

Prior Knowledge

Historical information provides the prior probability.

For example, if a road normally takes 30 minutes:

$P(\text{Time}=30 \mid \text{HistoricalData})$

can serve as prior information.

When real-time traffic data arrives, Bayesian inference updates the prediction.

Dynamic Updating

Suppose:

Before update:

$\text{ExpectedTime}(\text{RouteA})=30 \text{ min}$

A traffic accident is detected.

The system updates its belief:

$\text{ExpectedTime}(\text{RouteA})=55 \text{ min}$

It may then select Route B.

Probabilistic Route Selection

Instead of simply choosing the shortest physical distance, the system can minimize expected cost:

$$\text{ExpectedCost}(R) = P(\text{Traffic} | R) \times \text{TrafficCost} + P(\text{Weather} | R) \times \text{WeatherCost} + \text{DistanceCost}$$

The route with the lowest expected cost can be selected.

Computational Complexity

Traditional route algorithms such as Dijkstra's algorithm can efficiently find shortest paths, but dynamic Bayesian reasoning adds computational overhead.

For large transportation networks:

1. Many possible routes exist.
2. Traffic changes continuously.
3. Weather conditions change.
4. Predictions must be updated quickly.

Therefore, efficient algorithms and approximate inference may be required.

Effect of Uncertainty

Uncertainty can cause the system to avoid routes with high risk even if they are shorter.

For example:

Route	Distance	Traffic Risk	Expected Time
A	10 km	High	45 min
B	13 km	Low	30 min
C	8 km	Very High	60 min

The system would choose Route B, even though Route A/C may have shorter distances.

Conclusion

Bayesian inference allows the route optimizer to combine historical knowledge with real-time information. It is particularly useful when traffic and weather conditions are uncertain. The main challenge is maintaining fast computation while continuously updating predictions.

